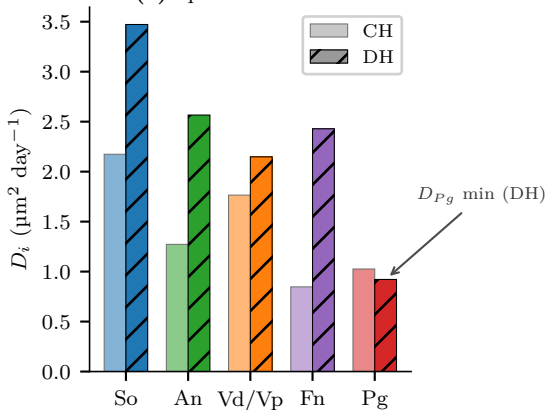
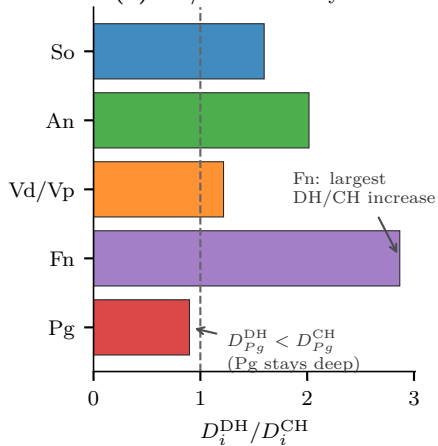


(a) Species diffusivities



(b) DH/CH diffusivity ratio



3D-PINN inverse fit (3000 epochs). CH loss=0.0046, DH loss=0.0060. $u_{\text{CH}}=-0.38$, $u_{\text{DH}}=+1.00 \mu\text{m day}^{-1}$.